(4) $z_1 = (a+2)+(b+1)i = (4)+(9)i$

modulus r_1

$\hookrightarrow r_1 = \sqrt{4^2+9^2} = \sqrt{16+81} = \sqrt{97} = 9{,}8489$

$\theta_1 = \arctan\left(\frac{9}{4}\right) = \arctan(2{,}25)$

$\theta_1 = 1{,}1526$ rad

$\theta_1 = 66{,}0375^0$

$z_2 = (b+1)-(a+2)i = 9-4i$

modulus r_2

$\hookrightarrow r_2 = \sqrt{9^2+(-4)^2} = \sqrt{81+16} = \sqrt{97} = 9{,}8489$

$\theta_2 = \arctan\left(\frac{-4}{9}\right) = -\arctan(0{,}4444$

$\theta_2 = -0{,}4189$ rad

$\theta_2 = -24{,}004^0$

$z_1 = 4+9i$

$z_2 = 9-4i$

* $z_1+z_2 = (4+9)+(9-4)i = 13+5i$

* $z_1 . z_2 = (4+9i)(9-4i) = 4\times 9+4\times(-4i)+9i\times 9+9i\times(-4i)$

$= 36-16i+81i-36i^2$

$i^2 = -1 \Rightarrow -36i^2 = 36$

$= 36+36+(-16i+81i) = 72+65i$

* $z_1/z_2 = \frac{4+9i}{9-4i} \times \frac{9+4i}{9+4i} = i$

* $|z_1 - z_2|$

$\hookrightarrow z_1 - z_2 = (4-9)+(9-(-4))i = -5+13i$

$|z_1-z_2| = \sqrt{(-5)^2+13^2}$

$= \sqrt{25+169} = \sqrt{194} = 13{,}9284$

Dipindai dengan CamScanner